

Part I

Graph Theory - I

Definitions and examples

Definition 1. A graph is a set of points, called vertices, together with a collection of lines, called edges, connecting some of the points. The set of vertices must not be empty.

If G is a graph we may write $V(G)$ and $E(G)$ for the set of vertices and the set of edges respectively. The number of vertices is written $|G|$, and the number of edges is written $e(G)$ (or $|E|$).

A graph may have multiple edges, i.e., more than one edge between some pair of vertices, or loops, i.e., edges from a vertex to itself. A graph without multiple edges or loops is called simple. Many natural problems only make sense in the setting of simple graphs.

For each vertex v , the set of vertices which are adjacent to v is called the neighbourhood of v , written $\Gamma(v)$. A neighbour of v is any element of the neighbourhood.

Remark 1. Normally we do not include v in $\Gamma(v)$; it is only included if there is a loop.

Definition 2. The degree of a vertex v , written $d(v)$, is the number of ends of edges which connect to that vertex.

Example 1. In a simple graph, $d(v) = |\Gamma(v)|$.

A loop: As both ends of a loop go to the same vertex, each loop contributes 2 to the degree.

Question 1. What happens if we add up the degrees of all the vertices?

Lemma 1 (Euler's Handshaking Lemma). *The sum of the degrees of the vertices of a graph is equal to twice the number of edges.*

Remark 2. We shall consider only locally-finite graphs, i.e., graphs such that vertices occur in finitely many edge pairs (each vertex is connected to only a finite number of edges).

Examples of graphs (Real World)

1. **Facebook graph:** V is the set of Facebook users and an edge is placed between two vertices if they are friends of each other.
2. **Network graph:** V is the set of cities and an edge represents roads/trains/air between cities.
3. **Collaboration graph:** V is the set of all mathematicians who have published articles and an edge represents that two mathematicians have collaborated on a paper together.

Definition 3 (Weighted graphs). A graph G with a weight function $W : E \rightarrow \mathbb{R}$.

Example 2. A road network where the weight on an edge denotes average traffic in a day.

Special Types of Graphs

Definition 4 (Complete Graph). A Complete graph is a graph in which every pair of distinct vertices is connected by an edge. If the graph has n vertices, the complete graph is denoted by K_n .

- K_n has exactly $\binom{n}{2} = \frac{n(n-1)}{2}$ edges.
- Every vertex in K_n has degree $n - 1$.

Definition 5 (Intersection Graph). Let $S_1, S_2, \dots, S_n$ be subsets of a set S . Let $\mathcal{F} = \{S_1, \dots, S_n\}$. Define a graph G as follows:

- Each vertex of G corresponds to a set $S_i \in \mathcal{F}$.
- Two vertices S_i and S_j are connected by an edge if and only if S_i and S_j intersect, i.e., $S_i \cap S_j \neq \emptyset$.

Lemma 2. *An intersection graph G is complete if and only if every two distinct sets in $\mathcal{F}$ intersect.*

Definition 6 (Delaunay Graph). Let $P \subseteq \mathbb{R}^N$ ($N \geq 1$) be a finite set of points. For $y \in \mathbb{R}^N$, $d(y, P) = \min_{x \in P} |x - y|$. Define the Voronoi cell of $x \in P$ as $C_x := \{y \in \mathbb{R}^N : d(y, P) = |x - y|\}$. This is the set of all points in $\mathbb{R}^N$ that are at least as close to x as to any other point in P . The Delaunay graph is the intersection graph on P with the Voronoi cells C_x as the intersecting sets. So, $G = (P, E)$ where $(x, x') \in E$ if and only if $C_x \cap C_{x'} \neq \emptyset$.

Definition 7. Euclidean Lattices: Let $B_r(x)$ be the closed ball of radius r centred at x . The n -dimensional integer lattice is the intersection graph formed with $\mathbb{Z}^n$ as vertex set and $B_T(z)$, $z \in \mathbb{Z}^n$ as the intersecting sets.

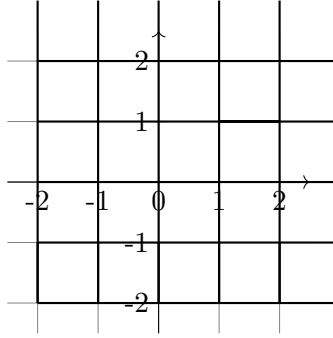
The N -dimensional integer lattice is the set $\mathbb{Z}^N = \{(n_1, n_2, \dots, n_N) : n_i \in \mathbb{Z}\}$.

Construction of the graph:

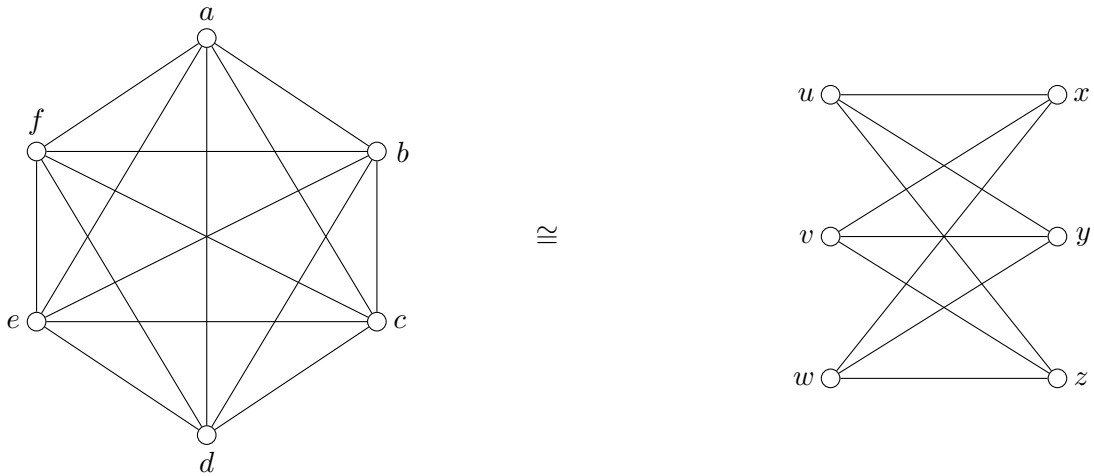
1. The vertex set is $V = \mathbb{Z}^N$.
2. For each lattice point $z \in \mathbb{Z}^N$, consider the closed ball of radius $1/2$ centered at z : $B_{1/2}(z) = \{y \in \mathbb{R}^N : |y - z| \leq 1/2\}$.
3. This is an intersection graph where the sets are these closed balls. Two vertices $z, z' \in \mathbb{Z}^N$ are adjacent if their closed balls intersect.
4. $(z, z') \in E \iff B_{1/2}(z) \cap B_{1/2}(z') \neq \emptyset$.

Characterization of Adjacency in Euclidean Lattice Graph

Two balls $B_{1/2}(z)$ and $B_{1/2}(z')$ intersect if and only if the distance between their centers is less than or equal to the sum of their radii: $|z - z'| \leq 1/2 + 1/2 = 1$. Since $z, z' \in \mathbb{Z}^N$, this condition means they differ in exactly one coordinate by ± 1 , and are equal in all others.



Definition 8 (Isomorphism). If G and H are two graphs, then an isomorphism between G and H is a bijection $\phi : V(G) \rightarrow V(H)$ such that for any $v, w \in V(G)$, the number of edges between v and w in G is the same as the number of edges between $\phi(v)$ and $\phi(w)$ in H . If such a bijection exists, we say that G and H are isomorphic and write $G \cong H$.



Definition 9. Let G and H be graphs. We say that H is a subgraph of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. If $V(H) = V(G)$, then H is a spanning subgraph. A proper subgraph is any subgraph other than the graph itself.

Definition 10. A graph is connected if it cannot be written as a disjoint union of two graphs. (This means there does not exist a partition of its vertex set $V = V_1 \cup V_2$ such that no edge connects a vertex in V_1 with a vertex in V_2).

Example 3 (Disconnected Graph). $V = \{1, 2, 3, 4\}$, $E = \{\{1, 2\}, \{3, 4\}\}$. No path exists between $\{1, 2\}$ and $\{3, 4\}$.

Definition 11 (Regular Graph). A regular graph is one in which all vertices have the same degree.

Definition 12. A graph is k -regular if every vertex has degree k .

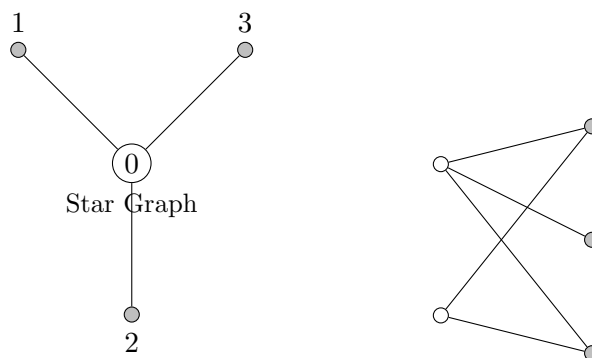
- A 0-regular graph is just a collection of isolated vertices.
- A 1-regular graph is a collection of disjoint edges.

In a k -regular graph with n vertices, the sum of degrees is nk . By the Handshaking Lemma, this is equal to $2|E|$. $\Rightarrow 2|E| = nk \Rightarrow |E| = \frac{nk}{2}$.

Definition 13 (Bipartite Graphs). A graph is bipartite if we can colour the vertices red and blue in such a way that each edge connects a red vertex to a blue vertex.

Alternatively, a bipartite graph's vertex set can be divided into two disjoint subsets V_1 and V_2 such that every edge of the graph joins a vertex in V_1 to a vertex in V_2 . No edge exists between two vertices within the same subset.

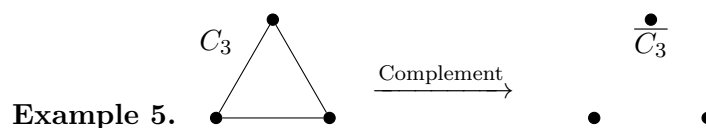
Formally, $G = (V, E)$ is bipartite if there exists a partition of the vertex set $V = V_1 \cup V_2$, $V_1 \cap V_2 = \emptyset$, such that $E \subseteq \{(u, v) \mid u \in V_1, v \in V_2\}$.



Example 4.

C_3 (a triangle) is not a bipartite graph. A hexagon (C_6) is bipartite.

Definition 14 (Complement Graph). Let G be a simple graph. The complement of G , written $\overline{G}$ or G^c , is the simple graph with the same vertex set as G such that two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G . The complement of the complement is the original graph, i.e., $\overline{(\overline{G})} = G$.



Example 5.

Walks, Paths, and Cycles

Definition 15. A walk in a graph is a sequence of the form $v_1, e_1, v_2, e_2, \dots, v_r$, for some $r \geq 1$, where v_i are vertices and e_i is an edge from v_i to v_{i+1} for each $1 \leq i < r$. (A walk is an alternating sequence of vertices and edges, starting and ending with vertices). Vertices and edges can be repeated.

Definition 16. A path is a walk in which all vertices are distinct (and hence all edges are distinct). A walk is closed if $v_r = v_1$.

Definition 17. A closed path is called a circuit. A circuit $v_1, \dots, v_{k-1}, v_1$ with no repetition of intermediate vertices (i.e., $v_1, \dots, v_{k-1}$ are distinct) is called a cycle.

Lemma 3. *Let G be a simple graph with at least two vertices. Then G must contain at least two vertices with the same degree.*

Trees

Theorem 1. *A graph G is connected if and only if there is a path between every pair of vertices.*

A graph has a cycle if it has a subgraph isomorphic to C_n for some n .

Definition 18. A tree is a connected graph with no cycles. A forest is a graph with no cycles (its components are trees).

Remark 3. A tree must be a simple graph. A vertex of degree 1 in a tree is called a leaf.

Theorem 2. *A connected graph with n vertices has at least $n - 1$ edges.*

Theorem 3. *Let G be a connected graph with n -vertices. Then G is a tree if and only if $e(G) = n - 1$.*

Recap of Walks, Trails, Paths, etc.

1. **Walk:** Edges and vertices can be repeated. E.g., $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow 3$.
2. **Trail:** Edges cannot be repeated, but vertices can be. E.g., $1 \rightarrow 3 \rightarrow 8 \rightarrow 6 \rightarrow 3 \rightarrow 2$.
3. **Circuit:** A closed trail (starts and ends at the same vertex, edges not repeated).
4. **Path:** Vertices and edges are not repeated.
5. **Cycle:** A closed path.

Theorem 4. *Every finite tree with at least two vertices contains at least two leaves (vertices of degree 1).*

Remark 4. For an infinite tree, this is not necessarily the case.

Euler's Theorem

Definition 19 (Eulerian Graph). A circuit (a closed trail) in a graph visiting every edge exactly once is called an Eulerian Circuit. A graph that has an Eulerian Circuit is called an Eulerian graph.

Theorem 5 (Characterization). *A finite connected graph is Eulerian if and only if every vertex has an even degree.*

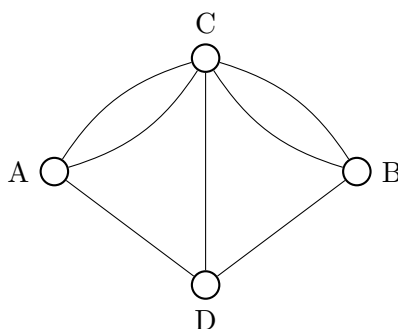
König's theorem on bipartite graphs

Theorem 6. *A graph is bipartite if and only if it has no odd cycles.*

Example 6 (The Königsberg Bridge Problem). The city of Königsberg was divided by the Pregel river into two islands and two mainlands, connected by seven bridges.

Question: Is it possible to start at some point, walk across every bridge exactly once, and return to the starting point?

Solution (Euler): Represent the land masses as vertices and bridges as edges. This creates a multigraph.



The degrees of the vertices are: $d(A) = 3$, $d(B) = 3$, $d(C) = 5$, $d(D) = 3$. Since not all vertices have an even degree, an Eulerian circuit is not possible.

Part II

Graph Theory - II (Notes)

Graphs as metric spaces

Let G be a connected graph. We can view graphs as metric spaces by defining the distance between two vertices v, w as the length of the shortest path between them.

$$d_G(v, w) := \inf\{\text{length}(P) \mid P \text{ is a path from } v \text{ to } w\}$$

Set $d_G(v, v) = 0$ for all $v \in V$, and $d_G(v, w) = d_G(w, v)$. (V, d_G) is a metric space.

Adjacency Matrices

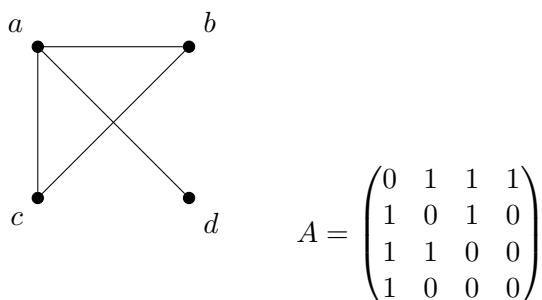
Suppose $G = (V, E)$ is a simple graph where $|V| = n$. Suppose the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$. The adjacency matrix A of G with respect to this listing is the $n \times n$ zero-one matrix with 1 as its (i, j) -th entry when v_i and v_j are adjacent, and 0 as its (i, j) -th entry when they are not adjacent.

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

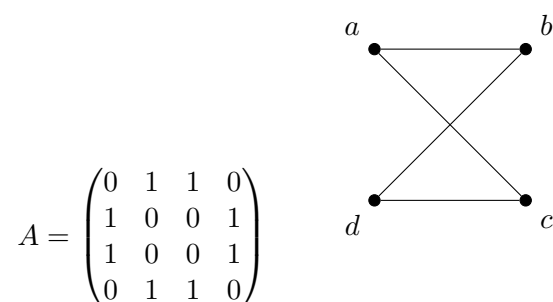
Properties:

1. The adjacency matrix depends on the ordering of the vertices. Hence, there can be up to $n!$ different adjacency matrices for a given graph.
2. The adjacency matrix of a simple graph is symmetric, i.e., $a_{ij} = a_{ji}$. It therefore has real eigenvalues.

Example 7. Use an adjacency matrix to represent the following graph with vertex ordering a, b, c, d .



Example 8. Draw a graph with the adjacency matrix below, with respect to the ordering of vertices a, b, c, d .



Lemma 4. Let G be a graph on n -vertices and A be its adjacency matrix. The entry $(A^k)_{ij}$ is the number of walks of length k from vertex i to vertex j .

Properties of Trees

Theorem 7 (Characterization of trees). *Let $G = (V, E)$ be a graph with n vertices and m edges. The following statements are equivalent:*

1. G is a tree (connected and acyclic).
2. There is a unique path between any two vertices in G .
3. G is connected, but $G \setminus \{e\}$ is disconnected for every edge e of G .
4. G is connected, and $m = n - 1$.
5. G is acyclic, and $m = n - 1$.
6. G is acyclic, but $G + xy$ has a cycle for every two non-adjacent vertices $x, y \in V$.

Definition 20 (Spanning Tree). A subgraph $T = (V_T, E_T)$ of a graph $G = (V, E)$ is a spanning tree if:

- (i) T is a tree, and
- (ii) $V_T = V$ (it is spanning).

Theorem 8. *A graph admits a spanning tree if and only if it is connected.*

Theorem 9 (Cayley's Formula). *The number of distinct spanning trees of a complete graph K_n is n^{n-2} for $n \geq 1$.*

Prüfer Code

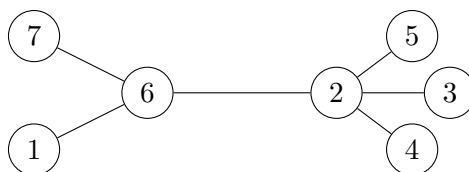
This formula can be proven by establishing a bijection between the set of spanning trees of K_n and the set of sequences of length $n - 2$ with elements from $\{1, 2, \dots, n\}$. This mapping is called the Prüfer code.

Algorithm to generate the code from a tree T : Repeat for $n - 2$ steps:

1. Find the leaf with the smallest integer label, call it s_i .
2. Let its unique neighbor be t_i .
3. The i -th element of the code is t_i .
4. Remove the vertex s_i and its incident edge from the tree.

This process stops when only two vertices remain.

Example 9 (Prüfer Code Generation). Let's find the Prüfer code for the following tree T :



- **Step 1:** Leaves are $\{1, 7, 3, 4, 5\}$. Smallest is 1. Its neighbor is 6. Code: (6). Remove 1.
- **Step 2:** Leaves are $\{7, 3, 4, 5\}$. Smallest is 3. Its neighbor is 2. Code: (6, 2). Remove 3.

- **Step 3:** Leaves are $\{7, 4, 5\}$. Smallest is 4. Its neighbor is 2. Code: $(6, 2, 2)$. Remove 4.
- **Step 4:** Leaves are $\{7, 5\}$. Smallest is 5. Its neighbor is 2. Code: $(6, 2, 2, 2)$. Remove 5.
- **Step 5:** Leaves are $\{7, 2\}$. Smallest is 2. Its neighbor is 6. Code: $(6, 2, 2, 2, 6)$. Remove 2.
- Stop. Two vertices $\{6, 7\}$ remain.

The final Prüfer code is $(6, 2, 2, 2, 6)$.

Decoding a Prüfer code is also possible, proving it is a bijection. A key property used is that the degree of a vertex k in the tree is equal to one plus the number of times k appears in the Prüfer code.

Minimum Spanning Tree (MST)

Definition 21. Let $G = (V, E)$ be a connected weighted graph with a cost/weight function $c(e)$ for each edge $e \in E$. The cost of a spanning tree $T = (V, E_T)$ is defined as $C(T) = \sum_{e \in E_T} c(e)$. A spanning tree T is a minimum spanning tree (MST) if $C(T) \leq C(T')$ for any other spanning tree T' of G .

Minimum Spanning Tree Algorithms

Two popular greedy algorithms for finding the MST are Kruskal's Algorithm and Prim's Algorithm.

Kruskal's Algorithm

1. Initialize a forest T with all vertices of G but no edges.
2. Create a list of all edges in G , sorted in non-decreasing order of their weights.
3. Iterate through the sorted edges. For each edge $\{u, v\}$:
4. If adding the edge to T does not form a cycle, add it to T .
5. Stop when the forest T becomes a single tree with $n - 1$ edges.

Prim's Algorithm

1. Initialize a tree T with a single arbitrary starting vertex from G .
2. Maintain a set of vertices V_T that are in the tree. Initially, this is just the starting vertex.
3. Repeat $n - 1$ times:
4. Find the edge $\{u, v\}$ with the minimum weight such that $u \in V_T$ and $v \notin V_T$.
5. Add this edge to the tree T and the vertex v to the set V_T .
6. The final tree T is an MST of G .

Theorem 10. *Prim's Algorithm produces a minimum spanning tree in a connected weighted graph.*